

# E-trikes for urban delivery: An empirical mixed-fleet simulation approach to assess city logistics sustainability

Mojdeh Azad<sup>a</sup>, William J. Rose<sup>b</sup>, John H. MacArthur<sup>c</sup>, Christopher R. Cherry<sup>a,\*</sup>

<sup>a</sup> Department of Civil and Environmental Engineering, University of Tennessee, Knoxville, TN 37996, United States

<sup>b</sup> Department of Supply Chain Management, Debbie and Jerry Ivy College of Business, Iowa State University, Ames, IA 50011, United States

<sup>c</sup> Transportation Research and Education Center, Portland State University, Portland, OR 97201, United States

## ARTICLE INFO

### Keywords:

Last-Mile Delivery  
(E-)cargo-cycle  
E-trike  
City Logistics  
Google API  
Sustainable Transportation

## ABSTRACT

Increasing urbanization has resulted in changes in land use, environment, demographics, and consumer behavior. These changes have magnified the challenges of city logistics. In particular, last-mile delivery freight has become a growing challenge for sustainable cities. Despite numerous studies on last-mile delivery solutions, little attention has been given to assessing the performance of e-cargo in mixed fleet systems. This study's contribution lies in its empirical monitoring and evaluation of e-trike performance for two months and parameterization of a citywide simulation. We demonstrated how GPS data coupled with publicly available data can be used to develop various performance indicators and provided comparison between real-world data and the Google Distance Matrix API. Furthermore, we estimated the costs if the company used vans (all or a mixed fleet) for the same e-trike deliveries. The costs associated with an e-trike-based delivery system are not dramatically higher than those associated with traditional methods. Labor costs were found to be the primary factor influencing cost trade-offs. However, emissions costs are reduced to a large extent supporting sustainability initiatives of the system. This proposed approach can be generalized to evaluate alternative vehicle types that simultaneously change vehicle form factor and performance and fuel types.

## 1. Introduction

For decades, urbanization has changed social and economic systems in cities worldwide (Sun, Chen, Li, & Huang, 2020). Changes in urban economies and infrastructures create significant transportation challenges (Madlener & Sunak, 2011), impacting both passenger transportation and city logistics. City logistics service providers encounter challenges across the triple bottom line (Elkington, 1998), including social concerns such as public health and safety, environmental issues related to emissions and fuel use, and economic impacts reflected in company profits and traffic congestion (Quak, 2008; Ranieri, Digiesi, Silvestri, & Roccotelli, 2018; Taniguchi, Thompson, & Yamada, 2016). Policies and measures that force trade-offs between social, environmental, and economic outcomes, along with system inefficiency and demand growth, hinder sustainability efforts in city logistics (Macharis & Kin, 2017). From an environmental perspective, city logistics accounted for only 3% of all freight activity in 2019, but contributed 20% of all freight emissions, which in turn represented 24% of global CO<sub>2</sub> emissions from fuel combustion (see Fig. 1) (International Energy

Agency, 2020; International Transport Forum, 2021).

Private companies can contribute to sustainability improvements, specifically the reduction or avoidance of harmful urban freight kilometers by decarbonizing urban delivery operations and shifting to alternative fuels. Light electric vehicles (LEVs) present an innovative and sustainable tool already used in cities around the world. In addition to environmental benefits, the use of light electric vehicles has challenged the size constraints for delivery vehicles and is particularly relevant in areas with limited infrastructure, where larger vehicles may face challenges navigating tight spaces or congested roads (Balm, Moolenburgh, Anand, & Ploos van Amstel, 2018; Iwan, Nürnberg, Jedliński, & Kijewska, 2021). In such contexts, the use of light electric vehicles has opened new possibilities for efficient, cost-effective, and sustainable delivery operations.

Several studies and projects view cargo cycles as a sustainable solution to urban freight emissions (Al Mashalah, Hassini, Gunasekaran, & Bhatt, 2022; Dolati Neghabadi, Evrard Samuel, & Espinouse, 2019; Ploos van Amstel et al., 2018; Wrighton & Reiter, 2016). Electric cargo (E-cargo) cycles, which combine the advantages of both LEVs and cargo

\* Corresponding Author: University of Tennessee, 321 John D. Tickle Building, 851 Neyland Drive, Knoxville, TN 37996.

E-mail address: [cherry@utk.edu](mailto:cherry@utk.edu) (C.R. Cherry).

<https://doi.org/10.1016/j.scs.2023.104641>

Received 21 February 2023; Received in revised form 5 May 2023; Accepted 9 May 2023

Available online 11 May 2023

2210-6707/© 2023 Elsevier Ltd. All rights reserved.

cycles, are particularly suitable for local food and perishable product deliveries with tight time constraints (Arnold, Cardenas, Sørensen, & Dewulf, 2018; Ploos van Amstel et al., 2018).

While several European studies examine (e-)cargo cycles, empirical e-trike delivery research in the United States remains limited. This paper provides an opportunity to advance the assessment and understanding of e-cargo cycles use in the last mile delivery within the United States by examining an e-trike delivery case study and demonstrating the potential benefits of Global Positioning System (GPS) data for city logistics stakeholders to (i) monitor e-trikes, (ii) compare e-trike performance with other modes, (iii) inform e-trike location decisions, (iv) evaluate and improve existing logistics policy, and (v) present data for performance analysis through simulation or other methods. We illustrate this approach through a case study of an established commercial e-trike operation in Portland, Oregon to measure operational performance. We then use those performance outcomes to calibrate a simulation model that can be used for policy and scenario analysis. This paper continues with a discussion of the background and literature review, followed by the methods section describing the case study, collection of data and resources, description of simulation, and analysis methods. Following a detailed description of the analysis and simulation results, a discussion of the method and findings is presented. Lastly, the paper closes with conclusion remarks and recommendations.

## 2. Background and Literature Review

Since 2011, the European Commission has funded a series of projects on the introduction, application, and assessment of (e-)cargo cycles in commercial and non-commercial settings. These studies often focus on (e-)cargo cycles' sustainability impacts in European cities (Dolati Neghabadi et al., 2019). Similar to traditional cargo cycles, e-cargo cycles have access to bicycle infrastructures, allowing operators to avoid urban road congestion. Additionally, e-cargo cycles produce less noise, emit fewer pollutants, and enjoy greater parking accessibility than vans or trucks, reducing circling and idling time and improving both social and environmental conditions (Sheth, Butrina, Goodchild, & McCormack, 2019).

Prior literature classifies cargo cycles using several factors, including number of wheels, single- or multi-track, and electric or non-electric assisted. Gruber (2021) separates cargo cycle types into six groups, illustrated in Fig. 2. Service providers use (e-) cargo cycles for mail and parcel delivery, point-to-point on-demand services, and last-mile delivery from a depot or hub. Additionally, providers may offer

bike-train-bike services, first-mile pickups, and advertising (Austrian Mobility Research, 2017). Payload limitations, capacity constraints, and certain mechanical aspects make each (e-)cargo cycle type suitable for different services. Weather, topography, and payload affect the performance of commercial cargo bicycles. A study of the performance of Long John commercial cargo bikes in six European capitals found that the amount of energy required for the rider to overcome wind can range from 12 to 23% of their theoretical capacity, and city hills can consume 13–34% of riders' theoretical energy (Giordano, Matthews, Baptista, & Fischbeck, 2022). As a result, service providers configure their cycle fleet to match the services offered. To determine whether to shift from more traditional modes to (e-) cargo cycles, providers consider the regulations specific to a service area, infrastructure and land use conditions along with real and potential urban consolidation centers or micro-hubs and potential customer base in an area.

Research on (e-)cargo cycles focuses mainly on economic and environmental outcomes, though some studies examine the social aspects of safety, quality of life, and public health through case studies and relevant data. For instance, e-cargo cycle employees often exhibit lower heart rates (de Mello Bandeira, Goes, Gonçalves, Márcio de Almeida, & de Oliveira, 2019). Furthermore, research associates (e-)cargo cycles with improved quality of life (Oliveira, Albergaria De Mello Bandeira, Vasconcelos Goes, Schmitz Gonçalves, & D'Agosto, 2017), increased employment opportunities (Sadhu, Tiwari, & Jain, 2014), more pleasant and accessible neighborhoods with less visual nuisance (Verlinde, Macharis, Milan, & Kin, 2014), time savings (Koning & Conway, 2016), and social acceptance (Navarro, Roca-Riu, Furió, & Estrada, 2016). Finally, smaller commercial vehicles, which include cargo cycles, reduce the risk of bodily harm and associate with fewer severe and fatal crashes than larger freight vehicles (Wang, Parajuli, Cherry, McDonald, & Lyons, 2022).

(E-)cargo cycles have the potential to replace up to half of motorized trips for goods, with commercial transport accounting for 1/3 of these trips (City Changer Cargo Bike, 2019; Wrighton & Reiter, 2016). Cargo cycles provide the greatest benefits when delivering smaller payloads in dense urban areas, including postal services and parcel and food delivery (Ploos van Amstel et al., 2018; Rudolph & Gruber, 2017). Several studies conclude trip efficiency comparable to vans when providers use cargo cycles for 10% to 48% of deliveries (Gruber, Ehrler, & Lenz, 2013; Lenz & Riehle, 2013; S. Melo & Baptista, 2017). Changing from vans to (e-)cargo cycles that can park on the sidewalk and remain close to the delivery points removes the costs and time required to access vehicle parking spaces or dedicated loading facilities (Sheth et al., 2019) and

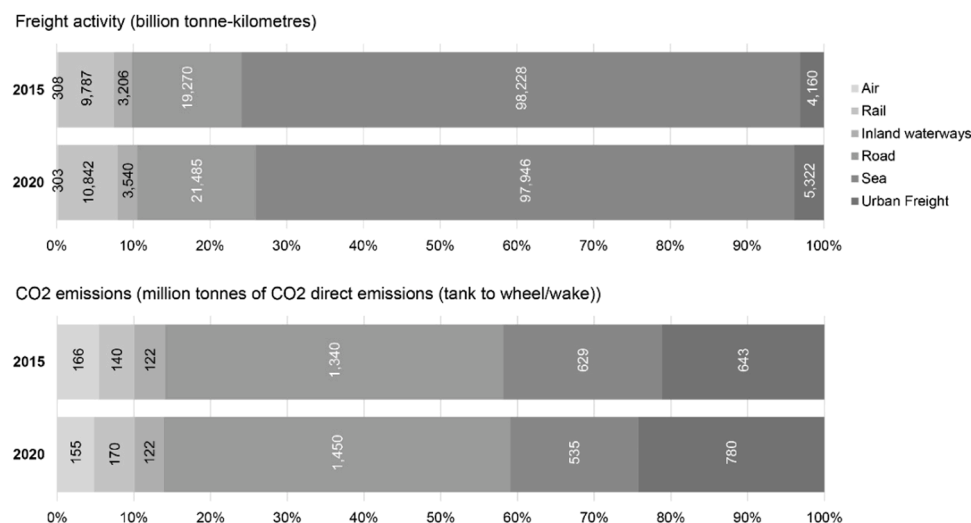


Fig. 1. Proportion of Freight activity and CO2 emissions by transport mode among six main modes (data adapted from the organization for economic co-operation and development (OECD) ITF Transport Outlook 2021 (International Transport Forum, 2021)).

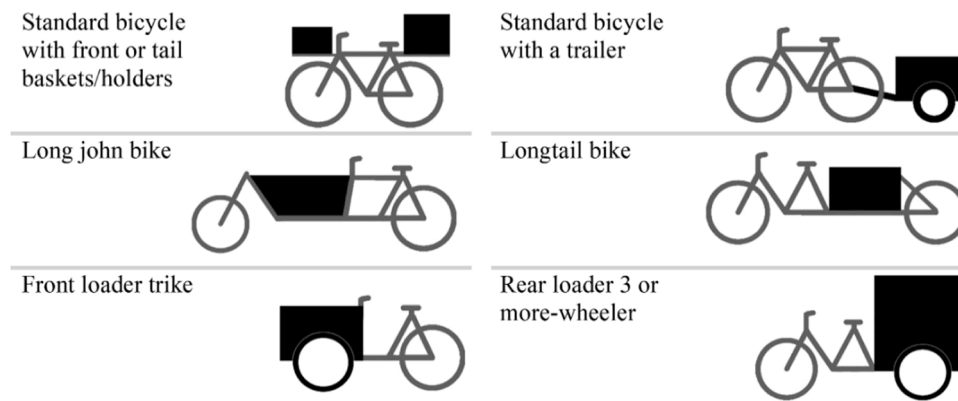


Fig. 2. (E-)cargo cycle classes.

combined with hubs, can result in increasing reductions of operating time and distance traveled as delivery density increases from low to 150 deliveries per km<sup>2</sup> (Dalla Chiara, Alho, Cheng, Ben-Akiva, & Cheah, 2020).

In addition to the costs associated with the operation (e.g., drivers), city policies, parking availability, and speed limits influence competition between (e-)cargo cycles and traditional vans (Jaller & Pahwa, 2021; Tipagornwong & Figliozi, 2014). These factors offer multiple cost reduction opportunities with (e-)cargo cycles in dense urban cores (Arnold et al., 2018; Elbert & Friedrich, 2020; Jaller & Pahwa, 2021; Lee, Chae, & Kim, 2019; Marujo et al., 2018; Sheth et al., 2019; Tipagornwong & Figliozi, 2014). Europe offers several examples of economically successful cycle logistics projects that achieve high profits and are favorable to start-ups when the bike model (i.e., trailer bike, cargo bike, tricycle, traditional bike) matches the urban and market factors (Giglio, Musmanno, & Palmieri, 2021).

E-cargo cycles also eliminate barriers associated with human-powered bikes, including grade changes, low average speeds, range limitations, or long-duration shifts. The above factors make e-cargo bikes a promising option for a wide range of urban logistics needs (Arnold et al., 2018; Ploos van Amstel et al., 2018). This class of LEVs has been piloted primarily by major parcel delivery companies such as UPS, DHL, and FedEx in Europe and in some locations in the U.S. (Lia, Nocerino, Bresciani, COLORNI VITALE, & Luè, 2014; Sheth et al., 2019).

Compared to delivery vans, e-cargo cycles reduce fuel consumption and carbon emission considerably, with research results ranging from 51% to 72% emission reductions (Assmann, Lang, Müller, & Schenk, 2020; Cairns & Sloman, 2019; Colson, 2019; Saenz, Figliozi, & Faulin, 2016; Tipagornwong & Figliozi, 2014). Furthermore, using the full share of cargo bikes in the fleet to deliver as many packages as possible is equivalent to replacing half of the diesel vans in the fleet with electric vehicles (Llorca & Moeckel, 2021). Using GPS data, another study associates cargo cycle delivery with positive environmental impacts in congested areas of New York City (Conway, Cheng, Kamga, & Wan, 2017). A Life Cycle Assessment study finds incorporating e-solar powered cargo cycles in Ghana could reduce greenhouse gas emissions by 4–8% (Schünemann, Finke, Severengiz, Schelte, & Gandhi, 2022). Cargo-bikes coupled with micro hubs can be effective solutions in the last echelon of multi-echelon strategies especially in dense areas, and when replacing diesel trucks, they can achieve marginal reductions of 0.62% reduction in emissions, 0.63% reduction in vehicle miles traveled, and 0.72% reduction in traffic accidents for every additional percentage of packages delivered by them (Pahwa & Jaller, 2022).

In summary, employing e-cargo cycles in city logistics enhances sustainability by lowering vehicle, maintenance, and parking costs as well as increasing travel speeds in dense urban cores, reducing driver training and certification requirements, and limiting adverse environmental impacts. At the same time, cargo cycle operations face several

challenges regarding security and safety concerns, capacity and range limitations, seasonality, inadequate infrastructure, stability, route scheduling, labor health, and labor cost (Behnke, 2019; Blazejewski, Sherriff, & Davies, 2020; Caggiani, Prencipe, Colovic, & Dell'Orco, 2020; Dybdalen & Ryeng, 2022; Katsela, Güneş, Fried, Goodchild, & Browne, 2022; Mayor of London, 2009).

In conclusion, studies show that (e-)cargo cycles have positive sustainability impacts and can provide a sustainable alternative to traditional delivery methods, particularly for smaller payloads in dense urban areas. While studies mainly focus on economic and environmental benefits, (e-)cargo cycles also offer social advantages, including improved quality of life, increased employment opportunities, and reduced risk of bodily harm. When planning for services, it is crucial to consider factors such as city policies, parking availability, and speed limits, along with infrastructure and land use conditions. The potential for (e-)cargo cycles to increase cost-effectiveness, decrease emissions and congestion, and improve social conditions highlights the importance of promoting the use of these vehicles in urban freight delivery.

To date, most studies rely on mathematical models to optimize networks and evaluate LEV cost-effectiveness for deliveries. Despite similar outcomes across mathematical models of costs or emissions, the magnitude of the effects differs (Gruber & Narayanan, 2019; Maes, 2017; Sheth et al., 2019). As many models utilize operator estimations instead of empirical data, measuring actual operations may provide additional clarity. However, empirical data on operational characteristics of e-trike-based delivery systems at the vehicle- and tour-levels remains limited.

This study investigates e-trike delivery operations in Portland, Oregon using GPS data. With 162 miles of bike lanes and 85 miles of bike paths, Portland represents a bike-friendly major city in the U.S. (Portland Bureau of Transportation PBOT, 2019). We choose to study the US-based company (B-Line: b-linepdx.com) based in Portland that has been using e-trikes for local deliveries since 2009. We instrumented the entire B-Line e-trike fleet (five vehicles) each day for two months and analyzed daily movements to identify real-world performance attributes. The data allowed us to develop speed, distance, and travel time profiles and uncover variations, identify stops, and calculate stop durations. We use these performance characteristics to develop parameters and build a simulation model using AnyLogic University Researcher, version 8.7.6, to compare e-trike performance against van use for last mile deliveries. Last-mile freight delivery is a continuing challenge for sustainable cities. An e-trike can contribute to the development of a last-mile delivery system that can help to address some of these difficulties. To date, very few studies have used empirical data to assess the performance of e-cargo in mixed fleet systems. Thus, we present one of the first studies to develop empirical estimates of e-trike performance and parameterize a citywide simulation model using real-world data.

### 3. Methods

To assess e-trike operational performance against conventional van performance, we developed approaches to characterize real-world (naturalistic) operations of an e-trike commercial delivery company. We collected GPS data to evaluate vehicle kinematic performance, delivery factors (stopped loading time), and tour characteristics (tour length, number of stops). We then compared and paired the GPS data with publicly sourced vehicle operations data from the Google API to generate a simulation framework assessing alternative fleet scenarios in Portland and investigate the performance of this service in other cities.

#### 3.1. Case study description

B-line is a certified B Corporation offering business-to-business delivery services with five e-trikes at the time of data collection (Chen & Kelly, 2015). The company operates seven days a week and offers several services including commercial food delivery (main service), product delivery, reverse logistics (recycling), advertising, and warehousing. They support sustainable local food producers by managing warehousing and delivery solutions. B-line is the longest running e-cargo delivery company in the US, long preceding more recent parcel delivery services. The company operates a single warehouse on the east side of the Willamette River, with most of their destinations in downtown Portland on the west side of the river.

The e-trike payloads range from 225 to 320 kilograms with cargo volumes between 1.6 to 2.2 cubic meter. E-trikes move at an average speed of 13 to 20 kilometers per hour and an average range of 12 to 24 kilometers per tour (Fig. 1). The maximum vehicle width ranges from 122 to 127 centimeters and the battery capacities are from 25 to 70 Amp-hours. While electrically assisted, the e-trike riders have to pedal to activate the e-trikes. They are considered e-bikes under Oregon law and retain the right to use bike lanes, bridges, and parking as regular two or three-wheeled e-bikes (MacArthur & Kobel, 2014).

#### 3.2. Data collection

We collected diverse sets of data to understand the performance of the system. Each e-trike carried a portable G-Log GPS Recorder during months of October and November 2017. The recorders collected location information, speed, and headings at 1 Hz resolution. To clean the data, we eliminated records with calculated speeds exceeding a defined threshold of 40 km/h or large discrepancies in calculated and recorded speed values. Table 1 presents a summary of collected data features. The description and characteristics of the data extracted from the GPS data are presented in the result section.

We define four main service area boundaries in Portland based on the GPS points (Fig. 3): 1) Northwest (NW): predominantly commercial, general employment, and high-density multi-dwelling zones, (2) Southwest (SW) including downtown and primarily commercial zones, (3) Northeast (NE) predominantly residential and some commercial zones, and (4) Southeast (SE) predominantly residential with some employment zones and the company warehouse.

**Table 1**  
Summary of GPS data.

| Delivery Characteristics    |                          |
|-----------------------------|--------------------------|
| Number of e-trikes          | 5                        |
| Number of GPS point records | 2.5 million              |
| Data resolution             | 1 point per second       |
| Collection period           | Oct 4, 2017- Dec 1, 2017 |
| Total number of tours       | 192                      |
| Total distance traveled     | 4500 km                  |

#### 3.3. Time/ speed comparisons and stop detection

To assess e-trike operational performance against vehicle travel time, publicly available data from Google APIs provided a basis for comparison. The GPS data included individual delivery points, determined using a manual process in ArcGIS for two weeks of data. Using consistent location information, departure times, months, and days, we obtained duration and distance values between consecutive stops (which we will call origin-destination or OD) for bicycles and motor vehicles using Google Distance Matrix API (R. A. Melo, Zarruk, Melo, & Imports RCurl, 2017). We obtained three estimates for driving: 1) best guess, 2) pessimistic, and 3) optimistic. The best guess model utilizes both historical traffic condition and live traffic, and the pessimistic estimates are generally longer and the optimistic estimates are shorter than the actual travel time, respectively.

Additionally, we used Movingpandas python library to identify the delivery points in our data (Graser, 2019; Graser & Dragaschnig, 2020). The algorithm would detect a delivery point when movements stay within a defined range longer than a pre-defined time. For this analysis, we used a minimum of 180 seconds and 50 meters as the thresholds to detect delivery points.

#### 3.4. Simulation and cost analysis

Next, we calculated the costs associated with replacing e-trike tours with cargo vans using AnyLogic simulation software. AnyLogic enables simulation and integration of multimethod behaviors and has been utilized in a variety of industries, including supply chain management, logistics, retail, manufacturing, market and competition analysis, business process, project and asset management, and traffic simulation (Ren, Zhao, Liu, & Chen, 2019; Zlatanovska et al., 2018).

One week of collected data provided simulation model parameters, including e-trike tours, delivery points and vehicle movement parameters (Table 2). The model ran for five simulated days, with 20 replications for each scenario. Upon completion of the initial run, results, total cost calculations, service levels, and total labor hours were analyzed against recommendations from Law, Kelton, and Kelton (2000) and 20 replications were determined to be adequate to achieve the accurate results. We simulated 16 potential fleet mix combinations to assess the feasibility of replacing multiple e-trike tours with a single cargo van (Table 3).

Following the routing order provided by the company, vehicles traveled through an entire tour in each replication. Throughout the tours, each vehicle followed a set activity sequence, starting at the firm depot, traveling to each delivery point, and serving the customer. Finally, vehicles return to the depot after either serving all customers or approaching a 13-hour time window. Fig. 4 shows the process followed by e-trikes.

To compare e-trikes and cargo vans, each mode utilized a specific transportation network. As vans travel on the road network, movement was determined using the shortest route recommendation provided by OpenStreetMap (available through AnyLogic). Travel speeds updated every 10 seconds and were determined using distributions associated with time of day, travel direction, and service area. All vans begin and end the tour at the firm's warehouse.

E-trikes enjoy greater flexibility than vans, with the right to ride in uncongested bike lanes and park on sidewalks, reducing the influence of traffic and distance from the warehouse to the delivery point. To model e-trike travel, GPS point data for five days was cleaned in ArcGIS to remove any points reflecting straight line travel by the e-trike. By measuring the angle between sets of three consecutive points in each sequence and removing points with a 180-degree angle, a set of GPS points corresponding to each change in travel direction was entered into AnyLogic and assigned to a specific e-trike agent. Each e-trike agent begins the tour at the tour's first GPS point and travels to each point in the sequence. On reaching a GPS point, e-trike agents updated speeds

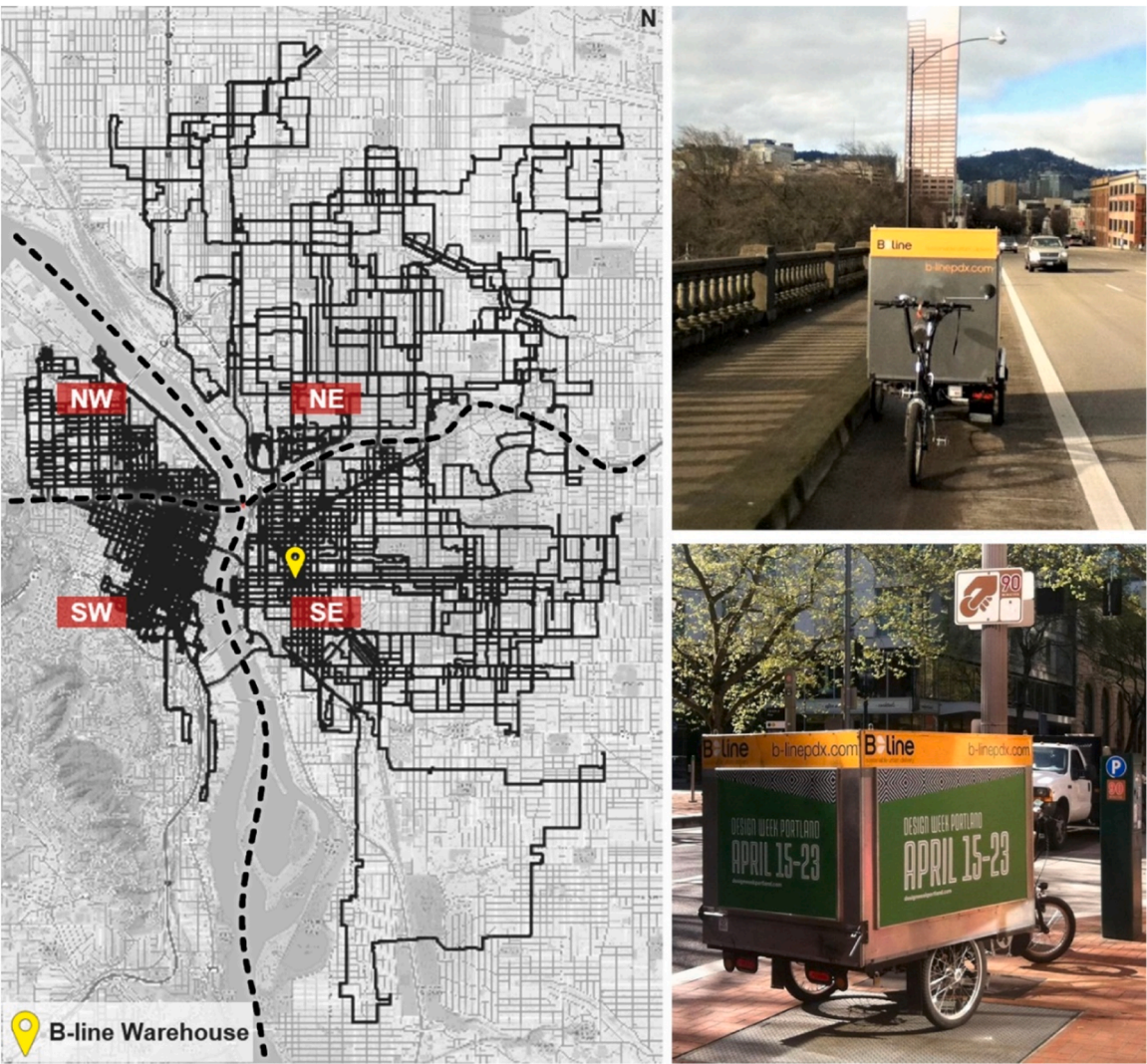


Fig. 3. Distribution of collected data in Portland and the defined boundaries (left), Delivery e-trikes: parking off-street (top right) and utilizing bike lane (bottom right) (images source: adapted from facebook.com/pg/blinedelivers).

Table 2  
Simulation parameters.

| Parameter         | Vehicle Type                             |                                                                                                             |
|-------------------|------------------------------------------|-------------------------------------------------------------------------------------------------------------|
|                   | E-Trike                                  | Diesel Van                                                                                                  |
| Driving Speed     | Varied depending on the time of day      | Varied depending on the time of day, location, and direction of travel                                      |
| Cruising Distance | 0                                        | Following a geometric distribution (p determined by the city “Centers & Corridors” parking study from 2015) |
| Service Time      | Derived from GPS data provided by B-line | Similar to e-trikes parameters                                                                              |
| Delivery Points   | Derived from GPS data provided by B-line | Similar to e-trikes parameters                                                                              |
| Fleet Mix         | 0–5                                      | 1–5                                                                                                         |

determined by distributions that varied by time of day and adjusted their heading to travel to the next GPS point. A separate dataset counted the number of GPS points between customers, with e-trikes traveling through the set number of GPS points before stopping at the pre-determined delivery point. Fig. 5 shows the GPS Points for all tours on Day 2 of the simulated week.

Upon reaching the delivery point, vans park in designated parking spaces, shared with other freight and passenger vehicles. To model the

Table 3  
Fleet mix options.

| Fleet Mix | Van | E-trike |
|-----------|-----|---------|
| 0         | 0   | 5       |
| 1         | 1   | 4       |
| 2         | 1   | 3       |
| 3         | 1   | 2       |
| 4         | 1   | 1       |
| 5         | 1   | 0       |
| 6         | 2   | 3       |
| 7         | 2   | 2       |
| 8         | 2   | 1       |
| 9         | 2   | 0       |
| 10        | 3   | 2       |
| 11        | 3   | 1       |
| 12        | 3   | 0       |
| 13        | 4   | 1       |
| 14        | 4   | 0       |
| 15        | 5   | 0       |

time spent cruising for an available parking space, the simulation modeled a Bernoulli trial, in which the number of attempts to find an open parking space follows a geometric distribution with a mean of  $1/p$  with  $p$  representing the percentage of total parking spaces open

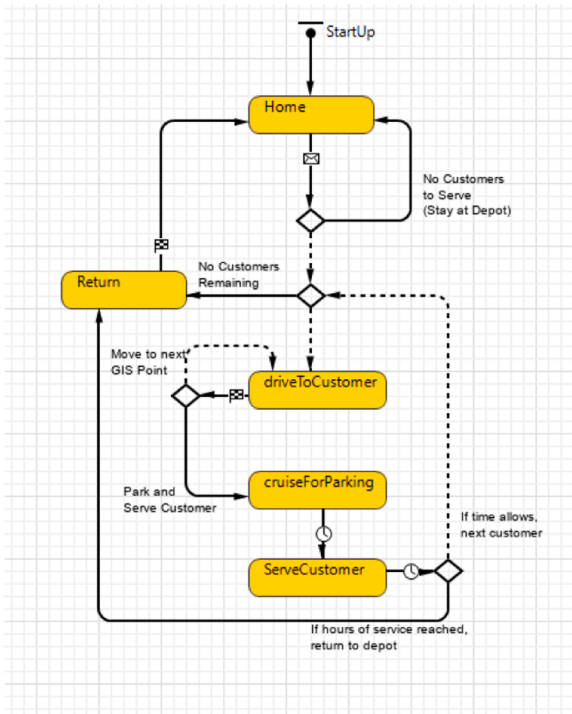


Fig. 4. Simulated process for e-trike travel.

(Geroliminis, 2015). Parking utilization data provided by the Portland Bureau of Transportation (Kittelson & Associates Inc., 2015) provided values for  $p$ . The data consists of both on-street and off-street parking supply and occupancy for five study areas, collected during typical mid-week days in 2015. Assuming a series of consecutive 16-foot long curbside parking spaces, cruising distance reflected the number of

attempts before success multiplied by 16, with cruising time determined by dividing the cruising distance by cruising speed. After parking, the time required to walk from the parked vehicle to the delivery point and back, at a speed of 3 miles per hour, was added to the service time provided by the firm. E-trikes, on the other hand, parked on sidewalks or streets near delivery points, making cruising for parking unnecessary and minimizing walking distance beyond service times. As a result, cruising time for e-trikes was assumed to be zero and customer service times were drawn from data provided by B-Line.

Cargo van tours were combined using either average distance per stop or the number of stops if the scenario modeled fewer than five vehicles. To maximize the potential of both vans and e-trikes, cargo vans served single tours with the longest average distance per stop. When a single van replaced multiple e-trike tours, the tours with the fewest delivery points were combined. To balance the workload, if multiple vans combined two tours, one van covered the tour with the most and fewest delivery points while another served the tours with the second and third fewest delivery points.

We modified Tipagornwong and Figliozzi (2014) cost function to account for attributes collected through GPS data. Discussions with LEV operators revealed vans as the most competitive alternative to e-trikes given the cargo characteristics and vehicle maneuverability and size. Further simulation experiments could assess additional vehicles and cost parameters. The following function calculated total cost in the simulation using the parameters presented in Table 4:

$$\text{Total Cost (per vehicle type)} = \text{energy cost} + \text{emission cost} + \text{maintenance cost}(c_m) + \text{labor cost}$$

Or

$$C_{tot} = [c_e](er) + [c_{CO2}][lerr_{CO2}] + [c_m](l) + [c_l](t) \quad (1)$$

Where:

$C_{tot}$  = total cost for vehicle type  $i$  (\$),  $c_e$  = unit energy cost for vehicle  $i$  [\$ /gal or \$ /kW-h],  $er$  = per mile fuel or electricity consumption rate of

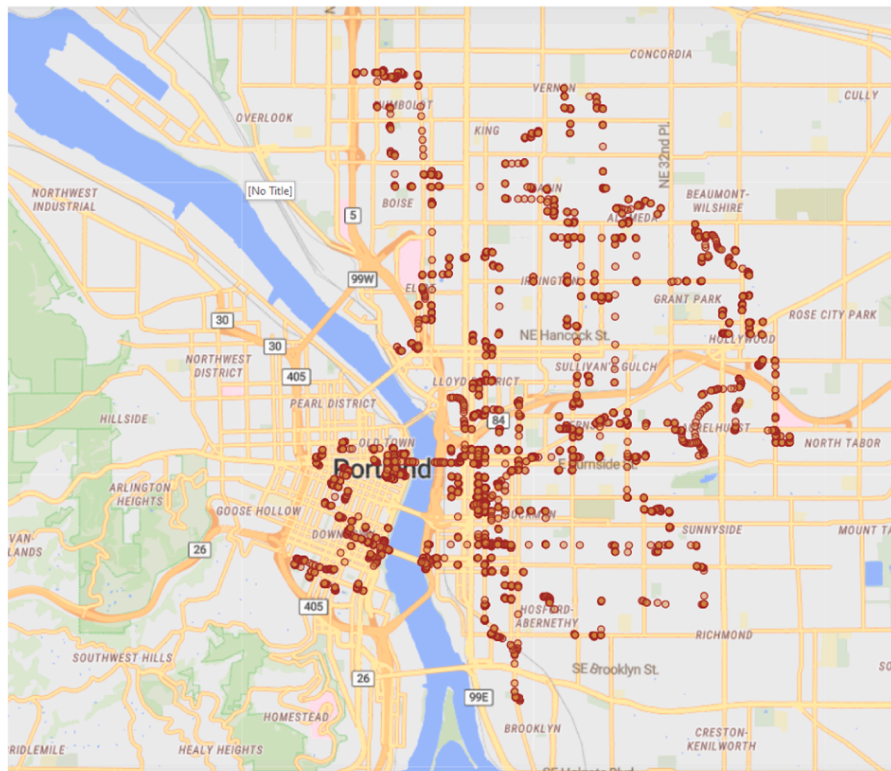


Fig. 5. GPS points for all tours (day 2).

**Table 4**  
Cost model parameters.

| Parameter     | Vehicle Type<br>E-trike | Diesel Van              | Source                                                         |
|---------------|-------------------------|-------------------------|----------------------------------------------------------------|
| Model         | Cycle<br>Maximus        | Dodge<br>ProMaster City | (Tipagornwong & Figliozzi, 2014)                               |
| $c_e$         | 8.45 cents/<br>kW-h     | \$2.993/gal             | (Electricity Local), (US Energy<br>Information Administration) |
| $c_m$ (\$/mi) | .02                     | .20                     | (Tipagornwong & Figliozzi, 2014)                               |
| $c_l$ (\$/h)  | 14,15,16                | 16.46                   | (B-Line), (Indeed.com)                                         |
| $c_{CO_2}$    | 18                      | 18                      | (Tipagornwong & Figliozzi,<br>2014), (Feng & Figliozzi, 2013)  |
| $E_r$         | 29 watt-h<br>mi         | 12–21 mi/gal            | (Tipagornwong & Figliozzi,<br>2014), (Kelley Blue Book)        |
| $r_{CO_2}$    | 0 kg/kW-h               | 10.180 kg/gal           | (Tipagornwong & Figliozzi,<br>2014), (SmartWay, 2018)          |
| L             | Simulation Output       |                         |                                                                |
| T             | Simulation Output       |                         |                                                                |

type  $i$  vehicles (gal/mi or kW-h/mi),  $c_{CO_2}$  = unit  $CO_2$  emissions cost (\$/ton), referenced to carbon taxes in the European cap and trade system as of 2013 (Feng & Figliozzi, 2013),  $l$  = per tour distance traveled to serve tour of vehicle  $i$  (mi/tour),  $r_{CO_2}$  =  $CO_2$  emission rate of vehicle type  $i$  (kg/gal or kg/kW-h),  $c_m$  = per mile maintenance cost for vehicle  $i$  (\$/mi),  $c_l$  = unit labor cost for vehicle  $i$  (\$/h), and  $t$  = total tour time of vehicle  $i$  (h).

(1 mile = 1.609 km, 1 US mpg = 235.215 L/100 km)

Reflecting company data, the simulation model set e-trike labor costs at \$14, \$15, and \$16 per hour. Van driver costs followed average delivery driver pay metrics from an online employment platform (Indeed.com). Online sources provided city fuel consumption for a Dodge ProMaster Cargo van and Dodge ProMaster City van at 15 and 21 miles per gallon, respectively. Therefore, the model set consumption rates at 15, 18, and 21 mpg along with the 12-mpg rate used in Tipagornwong and Figliozzi's study. With the Dodge ProMaster City's carrying capacity (131.7 cubic feet) of more than twice that of the average e-trike (67.2 cubic feet), fleet consolidation to less than three cargo vans would require the larger Dodge ProMaster Cargo van, with a carrying capacity of 309.5 cubic feet. Thus, the simulation used consumption and emission data for both van types. Diesel fuel costs for the week of October 9, 2017, provided fuel costs comparable to the e-trike GPS data collection period.

## 4. Results

This section presents the outcomes of this research study, which includes a descriptive analysis and simulation results by evaluating the costs associated with running a mixed fleet of e-trikes and vans. The descriptive analysis findings provide an in-depth assessment of the system's characteristics and performance, while the simulation results offer insight into the feasibility and effectiveness of the fleet mix by examining the various costs that are involved in operating a mixed fleet, such as total costs, labor costs, emission costs, and energy costs.

### 4.1. Descriptive analysis

Using the collected GPS data for two months, we first describe service characteristics. We removed 4.2% of GPS points as noise and outliers based on cleaning criteria and summarized the travel time, distance, and speed profiles using the remaining points. Delivery speed represents a primary economic factor influencing cost estimation and performance of a delivery service. Combined with perishable product time-constraints, speed impacts are critical. We calculated an average moving speed of 13.2 km/h ( $SD = 4.75$ ), with minimum movement speeds set at 2 km/h, average daily travel distance of 23.5 km per e-trike ( $SD = 8.44$ ), and daily tour times averaging 3.57 h per e-trike ( $SD = 1.42$ ) (Fig. 6). Additionally, the results indicate disparities in average speeds between different days and hours, as well as between multiple

infrastructures and areas. Accounting for speed variations on different roads and neighborhoods when routing and scheduling operations can enhance system performance. Collecting additional on-site information during data collection also contributes to understanding and better predictions for the system in the future. Note that all descriptive statistics emerged from GPS data collected during a two-month period in Portland, thus a number of factors could influence the estimates in other markets.

Further, the GPS data and Movingpandas python library estimated all delivery points based on stop duration. Using 180 seconds as the minimum threshold to define a delivery points, the results may place delivery points at intersections or locations with tour delays beyond three minutes. Extending the minimum threshold to 240 seconds, results show an average of 12 delivery points per e-trike ( $SD = 6$ ) with individual tours including 1 to 30 delivery points. Fig. 7 illustrates the distribution of delivery points detected under the two criteria. The number of delivery points varies significantly by the time of the day. As illustrated in Fig. 8, most delivery points require deliveries in the early morning, followed by the morning to noon segment. However, delivery point service times remain consistent throughout the day, indicating similar walking distance and loading and unloading times for all delivery points.

Following the initial simulation, we retrieved Google estimated travel times and distances for 1124 OD pairs from two weeks of collected GPS data. Fig. 9 illustrates the total number of delivery trips collected via GPS from Oct 4, 2017 (Wednesday) to Oct 17, 2017. Most deliveries occurred mid-week. Observed travel times for all OD pairs averaged 5 minutes with a 4.3 minute median, but 3.7 minutes when excluding trips to or from the depot (Fig. 10). Compared to estimated travel time by car for the same OD, e-trikes offered greater benefits than expected at shorter distances. However, relative travel time difference, defined as the travel time difference e-trikes and motor vehicles divided by e-trike travel times, ranges from -20% in optimistic car travel time values to 16% in pessimistic and is less competitive with e-trikes in shorter distances (Fig. 11). Therefore, the location of urban consolidation centers or micro-hubs significantly influences local delivery performance (Dalla Chiara et al., 2020; Robichet, Nierat, & Combes, ; Rudolph, Nsamzin-shuti, Bonsu, Ndiaye, & Rigo, 2022). Cities should also be involved in choosing sites and connecting private and public stakeholders in this aspect. In addition, the main stakeholder groups (courier express and parcel services, city administrations, and residents) were found to share environmental concerns and goals for the establishment of cooperative urban logistics hubs. Urban logistics hubs can be optimized to maximize the overall utility of the space across different users by maximizing the overall utility of the space (Russo, Voegl, & Hirsch, 2021).

### 4.2. Simulation results

With consistent fleets of only e-trikes or only vans, labor cost drives cost differences in the model, as expected. When e-trike drivers receive \$14 per hour, e-trikes provide more cost-effective solutions than vans. Increasing the pay of the e-trike driver, however, results in a greater total cost for e-trikes than vans. Table 5 presents the comparison of costs between a scenario with 5 e-trikes and van-only scenarios with 100% service levels. The total cost for the vans-only scenarios over the e-trike fleet varied between -9.4% to +5.9%, with average of 17% reduction in time, 2%-15% reduction in labor costs, and 5300%-9500% increase in energy costs. Note that in our OD comparisons we also found that the time deviation of car trips from e-trikes had an average between 16% (pessimistic) to -20% (optimistic), and -3% (pessimistic) to -30% (optimistic) when considering trips within  $\pm 100\%$  time deviations.

Incorporating mixed fleets in the simulation adds a layer of complexity. Variations in fuel consumption and driver pay altered scenario results. Additionally, the service level metric became increasingly important when fleets included fewer vans. As expected, running fewer vehicles overall reduced total costs. Meanwhile, as vehicles were given

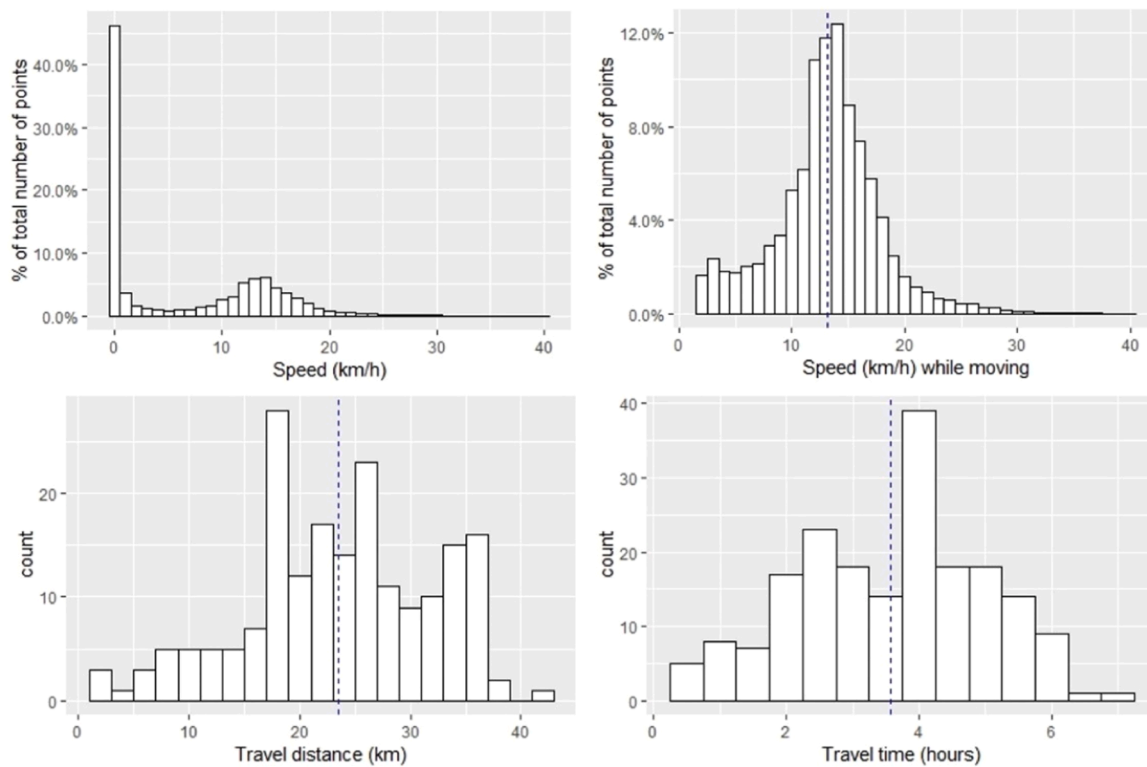


Fig. 6. Speed, travel distance, and travel time per e-trike per day.

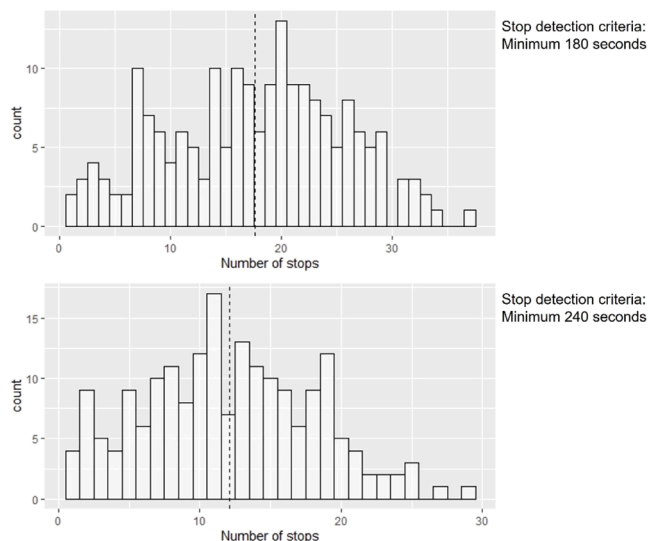


Fig. 7. Number of stops with the minimum of 180 seconds (top) and 240 seconds (bottom).

13 hours to complete their deliveries, tightening time windows may provide even more dramatic service level declines and alter the cost equation.

We found single van fleets with one or zero e-trikes failed to serve all delivery points during the proposed time period with service levels of 87% and 50%, respectively. With a 95% minimum service level, the results indicate that adding an e-trike increased average costs in these scenarios by 7% (three e-trikes) and 3% (four e-trikes), with cost disparities tied to e-trike rider hourly pay. The total tour time increased by 9% and 6% and labor costs increased by 8% and 4% with the third and fourth e-trike in the fleet, respectively. However, the average energy

cost decreased by 34% to 60% when adding third and fourth e-trike. A detailed presentation of the total cost and energy cost are illustrated in Figs. 12–14.

The range of additional average energy costs incurred by the fleet varies more extensively when e-trikes are absent, compared to scenarios where e-trikes are included in the fleet. In the latter case, the differences in energy costs are comparatively narrower. This finding implies that the absence of e-trikes from the fleet may lead to a more unpredictable range of energy costs depending on other factors related to the delivery. In contrast, when e-trikes are included in the fleet, the range of variation in energy cost is comparatively narrower. This finding underscores the importance of incorporating e-trikes into last-mile delivery operations as a means of promoting consistency and predictability in energy costs. Similar to energy costs, the range of variation in emission costs is more extensive when e-trikes are absent from the fleet, and the inclusion of e-trikes can help mitigate this variability.

As the labor cost is the primary driver of the total cost of delivery operations, the presence or absence of e-trikes in the fleet influences the total cost structure. Specifically, the analysis of the simulation scenarios revealed that the range of total cost is wider when e-trikes are present in the fleet, as compared to scenarios where e-trikes are absent, depending on various factors such as the routes and number of e-trikes being used in the fleet. Adding the first e-trike to the two van fleet increased the average cost by 7% with no additional costs for adding a second or third e-trike. The average energy cost decreased by 22%, 28%, and 39% when adding one, two, and three e-trikes to the fleet, with the average total time increase of 12%, 3%, and 4%, respectively. The results did not follow the same pattern in a fleet of three vans. Adding one e-trike to a fleet of three vans decreased the average total cost by 6% while also resulting in a decrease in energy cost by 31% and a decrease in total time by 2%.

Generally, fleet mix showed no significant influence on total cost associated at 100% service levels. The significant cost differences emerged from scenarios with zero e-trikes or a fleet of one van and two e-trikes, three vans and one e-trike, and four or five e-trikes with one

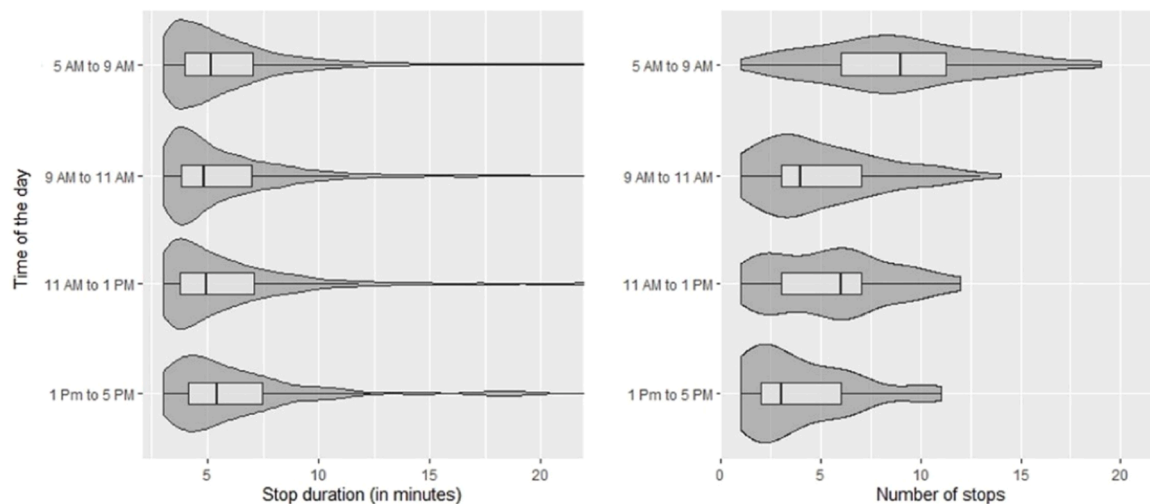


Fig. 8. Stop duration (left) and number of stops (right) by time of the day.

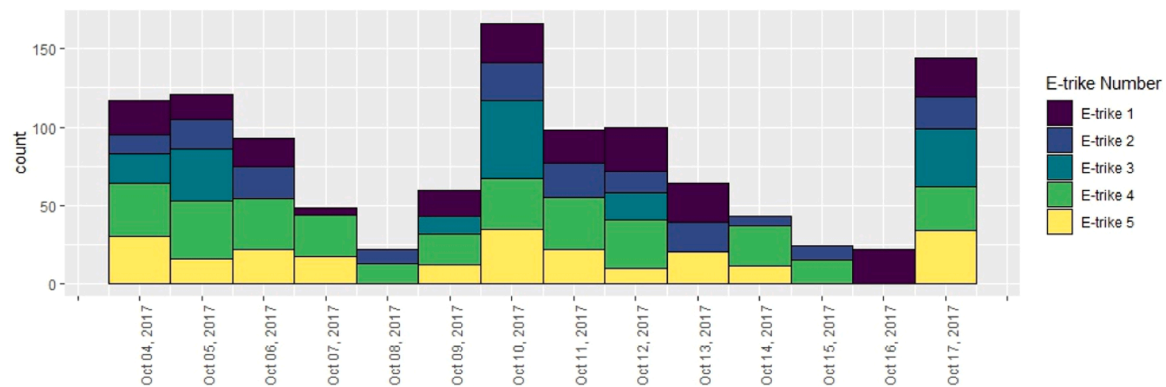


Fig. 9. Distribution of trips during the selected weeks.

van. Using the most efficient cargo vans, total cost differences were less than \$100 over the course of one week. However, the energy cost differed significantly between most scenarios unless fleets included one or zero e-trike and two to four vans.

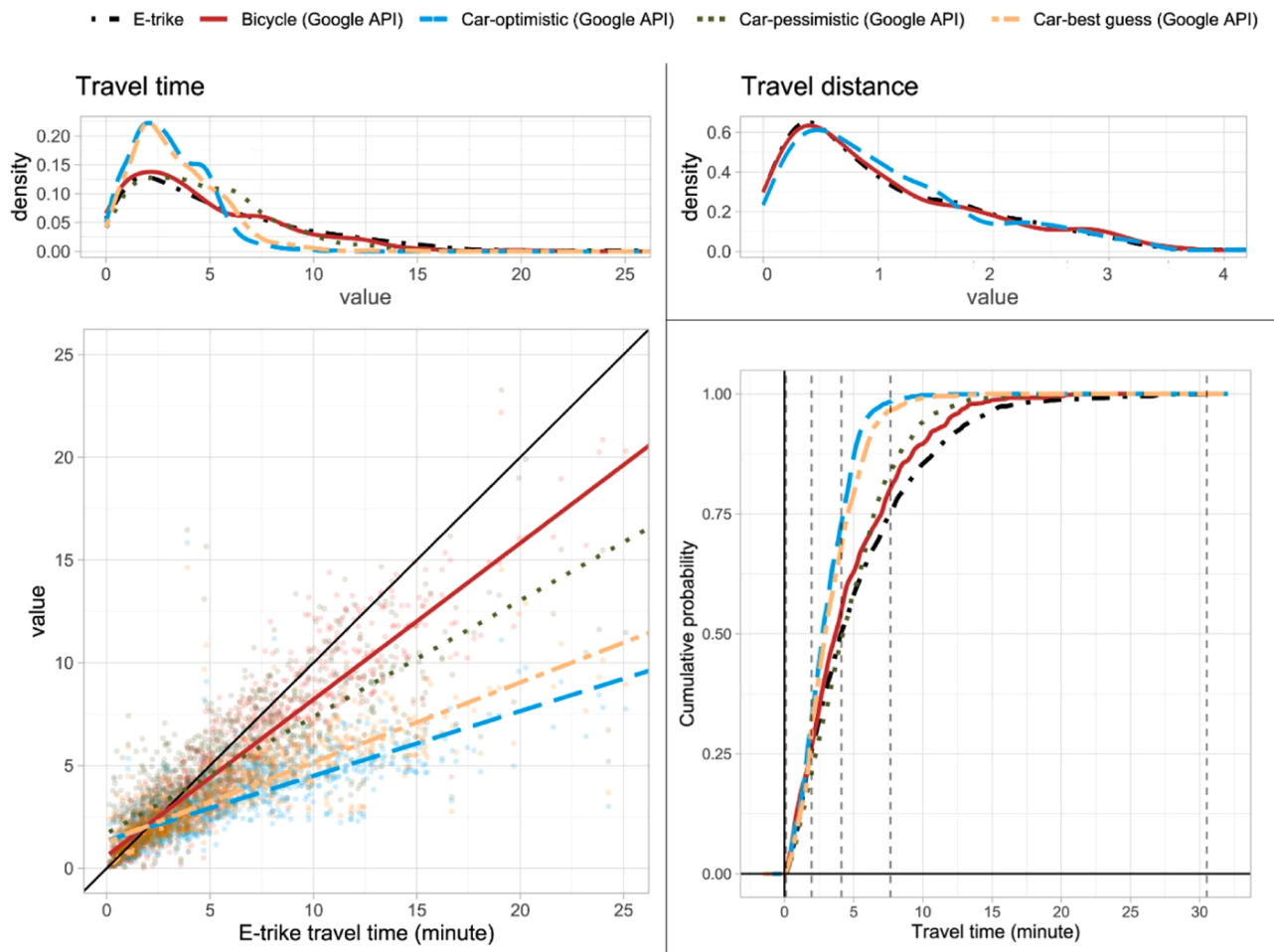
## 5. Discussion

Prior research has focused on characterizing and analyzing (e-)cargo cycles, primarily within European nations. In contrast, the present study represents one of the first attempts to empirically monitor a service for two months and parameterize a citywide simulation relying on precise locations and detailed transportation-related information to better understand the impact of e-trike on last mile delivery operations. A multi-method approach was used to accomplish the research objectives. First, GPS data was collected from e-trikes for two months to analyze their travel patterns. Second, detailed analysis was conducted to determine characteristics of each delivery route such as travel speed and duration. Third, a comparison was made between e-trike real-world data and travel time/distance for other modes retrieved from the Google Distance Matrix API. Finally, a citywide simulation was developed to analyze fleet mix and associated costs and evaluate the potential impact of e-trike usage on last mile delivery operations.

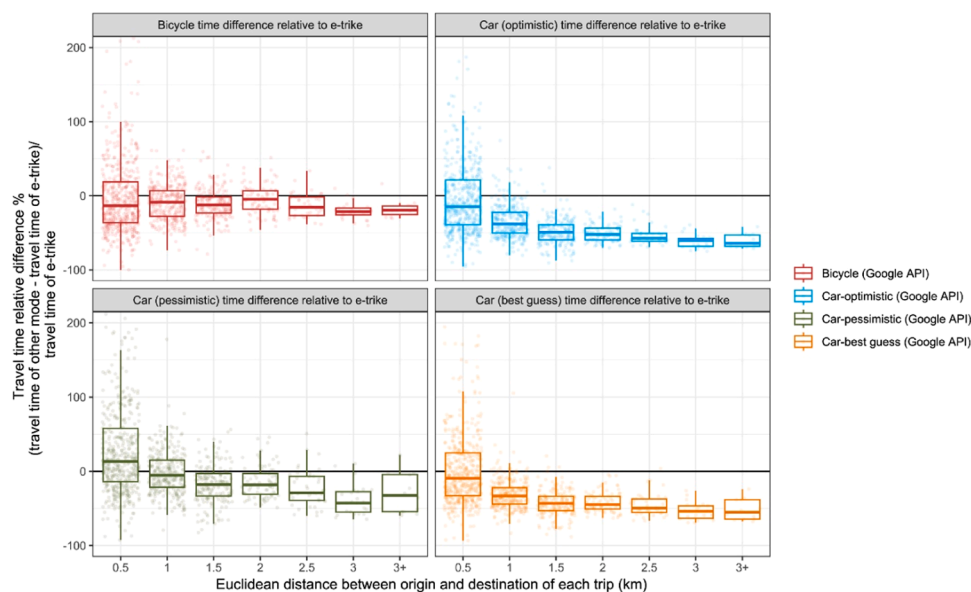
One of the main goals of this study was to closely monitor the service and demonstrate the efficacy of utilizing GPS data in a citywide simulation. This approach was combined with the inclusion of route and traffic/parking information from the city to enable cost analysis. The motivation for this approach was to provide a more accurate depiction

of the real-world situation and enable more robust cost assessments, without detailed operational data. To this end, this study provides a different methodology from previous studies that had used hypothetical routes or neglected traffic/parking input. By doing so, this study aimed to highlight the importance of incorporating real-world data in cost analysis, and the potential consequences of overlooking this crucial aspect. Therefore, this study underscores the significance of using real-world data when making informed decisions about city policies by considering the impacts of those policies on other stakeholders. Moreover, the proposed methodology can be applied more broadly to evaluate the efficacy of different vehicle types that may involve changes to vehicle form factor, performance, and fuel type, among other factors.

The results of the simulation have also been in line with prior research, indicating that the replacement of e-cargo cycles with diesel vans in a fleet would have adverse effects on the environment, energy consumption, and emissions costs across various fleet compositions. Prior research using the same service provider has indicated that the competitiveness of e-trikes was significantly influenced by drivers' expenses, while the impact of electricity or diesel costs was minimal (Tipagornwong & Figliozzi, 2014). In addition, the utilization of e-trikes was found to have the potential to reduce CO<sub>2</sub>e emissions by 50% to 70% (Saenz et al., 2016). However, this study took several additional steps to the prior studies by exploring the combinations of e-trikes and vans in the fleet, relying on precise customer location and the routes used by the provider in real-world, and incorporating more detailed transportation-related parameters in the simulation regarding speed and parking supply distributions in different city boundaries. In sum, the



**Fig. 10.** OD's Travel time density by mode (top left), travel time from Google API vs E-trike (bottom left), travel distance density by mode (top right), and travel time cumulative probabilities of e-trike and Google API for different modes (bottom right).



**Fig. 11.** Travel time relative differences by travel distance between OD's.

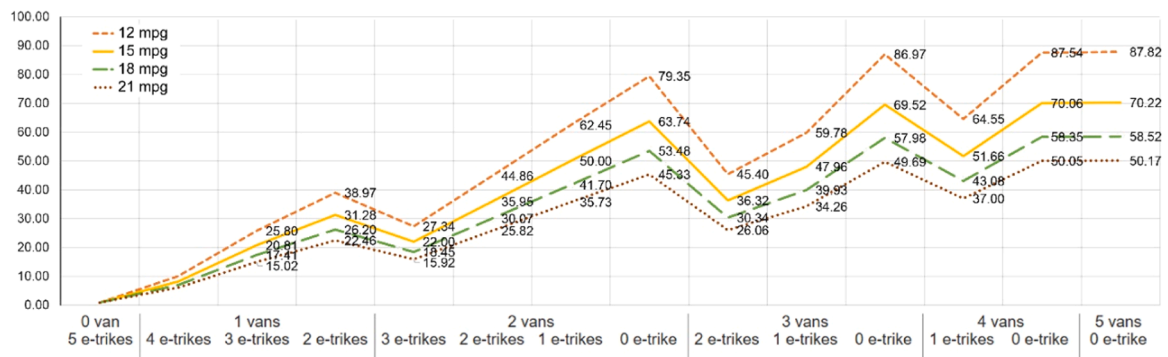
findings of this study lend support to the notion that employing e-trike cycles in last-mile delivery operations can have a significant positive impact on environmental outcomes, as evidenced by the reduction of

emissions costs and related environmental impacts. These outcomes are highly favorable in the context of broader sustainability initiatives aimed at enhancing the environmental performance of the logistics

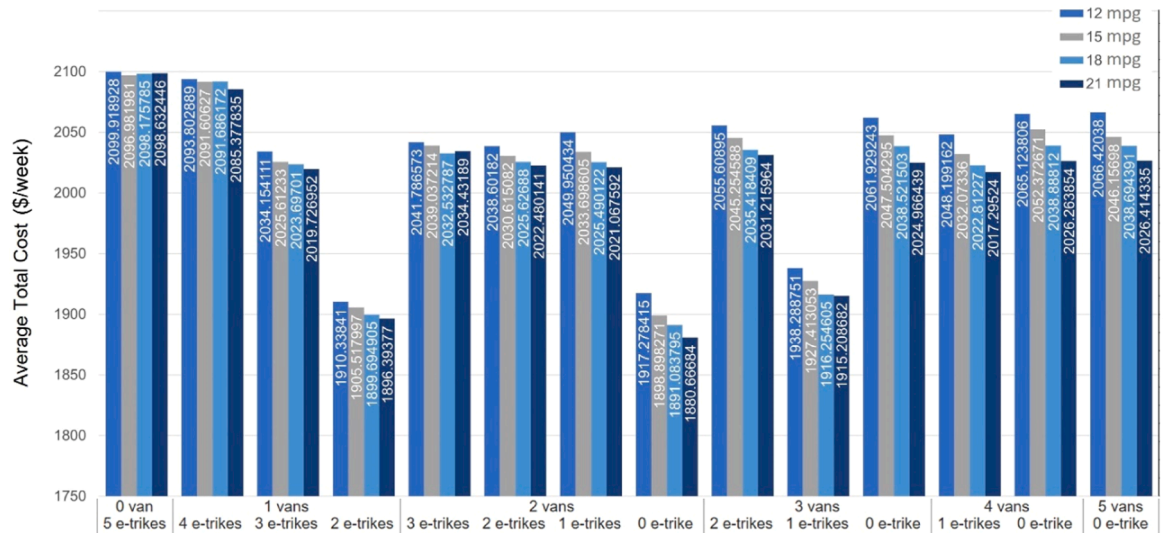
**Table 5**

Cost and time comparison between 5 e-trikes-only scenarios with van-only scenarios with service level of 100%.

| Scenario | e-trike pay (\$/h) <br>Consumption (mpg) | Average of<br>Total Cost | Average of<br>Energy Cost | Average of<br>Labor Cost | Average of Total<br>Time | Changes compared to the fleet with only 5 e-trikes |                        |                        |
|----------|------------------------------------------|--------------------------|---------------------------|--------------------------|--------------------------|----------------------------------------------------|------------------------|------------------------|
|          |                                          |                          |                           |                          |                          | Total cost<br>change %                             | Labor cost<br>change % | Total time<br>change % |
| 0 van    | \$14/h                                   | 1955.82                  | 0.92                      | 1947.42                  | 139.10                   |                                                    |                        |                        |
| 5 e-     | \$15/h                                   | 2098.43                  | 0.92                      | 2090.03                  | 139.34                   |                                                    |                        |                        |
| trikes   | \$16/h                                   | 2235.53                  | 0.92                      | 2227.13                  | 139.20                   |                                                    |                        |                        |
| 3 vans   | 12 mpg                                   | 2066.01                  | 86.96                     | 1903.45                  | 116.20                   | -7.6 to 5.6                                        | -14.5 to -2.3          | -16.6 to -16.5         |
| 0 e-     | 15 mpg                                   | 2048.28                  | 69.57                     | 1904.29                  | 116.26                   | -8.4 to 4.7                                        | -14.5 to -2.2          | -16.6 to -16.4         |
| trike    | 18 mpg                                   | 2036.27                  | 57.97                     | 1904.66                  | 116.27                   | -8.9 to 4.1                                        | -14.5 to -2.2          | -16.6 to -16.4         |
|          | 21 mpg                                   | 2025.02                  | 49.71                     | 1902.18                  | 116.12                   | -9.4 to 3.5                                        | -14.6 to -2.3          | -16.7 to -16.5         |
| 4 vans   | 12 mpg                                   | 2068.52                  | 87.57                     | 1904.82                  | 115.88                   | -7.5 to 5.8                                        | -14.5 to -2.2          | -16.8 to -16.7         |
| 0 e-     | 15 mpg                                   | 2050.47                  | 70.06                     | 1905.47                  | 115.93                   | -8.3 to 4.8                                        | -14.4 to -2.2          | -16.8 to -16.7         |
| trike    | 18 mpg                                   | 2039.07                  | 58.34                     | 1906.61                  | 116.00                   | -8.8 to 4.3                                        | -14.4 to -2.1          | -16.8 to -16.6         |
|          | 21 mpg                                   | 2027.14                  | 50.06                     | 1903.46                  | 115.80                   | -9.3 to 3.6                                        | -14.5 to -2.3          | -16.9 to -16.8         |
| 5 vans   | 12 mpg                                   | 2071.63                  | 87.81                     | 1907.48                  | 115.89                   | -7.3 to 5.9                                        | -14.4 to -2.1          | -16.8 to -16.7         |
| 0 e-     | 15 mpg                                   | 2047.61                  | 70.24                     | 1902.23                  | 115.57                   | -8.4 to 4.7                                        | -14.6 to -2.3          | -17.1 to -16.9         |
| trike    | 18 mpg                                   | 2038.77                  | 58.52                     | 1905.92                  | 115.79                   | -8.8 to 4.2                                        | -14.4 to -2.1          | -16.9 to -16.8         |
|          | 21 mpg                                   | 2025.91                  | 50.17                     | 1901.96                  | 115.55                   | -9.4 to 3.6                                        | -14.6 to -2.3          | -17.1 to -16.9         |



**Fig. 12.** Average energy cost for scenarios with service level of 95%+ by van fleet size.



**Fig. 13.** Total cost for scenarios with service level of 95%+ by van fleet size.

system.

## 6. Conclusions and recommendations

Recently, electric-assisted (e-)cargo cycles have emerged as a sustainable and innovative city logistics solution. Despite this, a lack of real-world data has limited performance analysis of emergent systems at

a micro-scale. This study contributes to understanding the e-cargo cycle use in last-mile delivery within the United States by measuring e-trike performance with naturalistic data and analyzing e-trike impacts on urban freight delivery systems. We used several methods to analyze the GPS data collected from a Portland-based delivery company over two months. Combining GPS and Google API data, a simulation model evaluated system performance. Analysis compared multiple modes

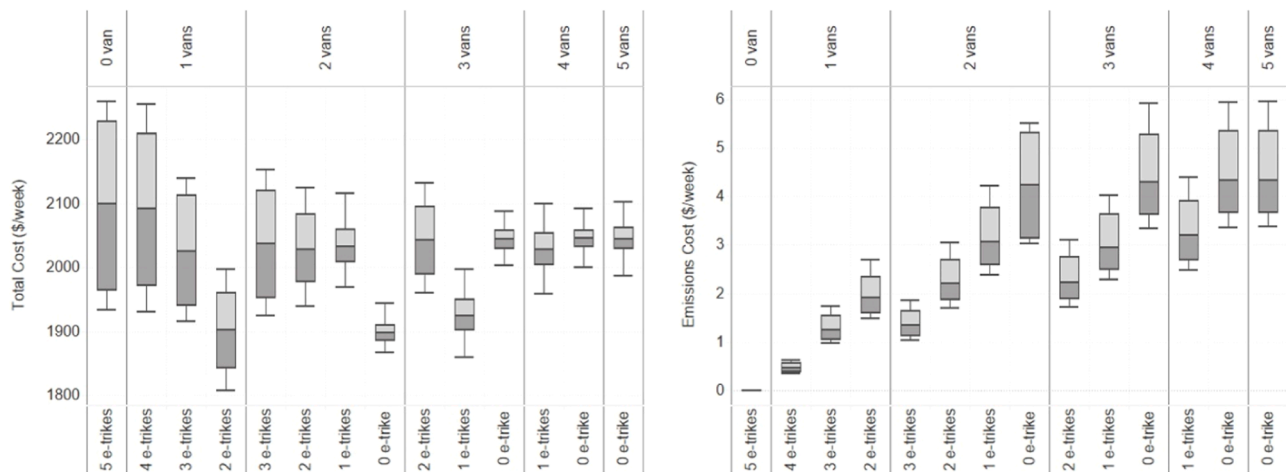


Fig. 14. Total cost and Emission cost for scenarios with service level of 95% or more.

using travel speeds and delivery point characteristics. Multiple factors, including traffic conditions and infrastructure, affected system performance at different levels and therefore should be incorporated in cost and feasibility evaluations when implementing e-cargo cycles for last-mile delivery.

To overcome the data gap, we recommend development of a framework to guide and enhance data generation and collection using multiple sources (e.g., rider/e-trike travel logs, GPS, phone applications). GPS data collection and sharing can provide valuable insights into e-trike performance and impacts for various stakeholders. Such information can improve routing, policies, and business plans to maximize opportunities to achieve numerous business and public sector objectives. Municipalities should collaborate with other public and private organizations and evaluate and improve policies to enhance citizen safety, reduce congestion in urban areas, and decrease motor vehicle emissions and noise.

We also propose a framework to analyze system costs applicable for multiple cities. For example, computational tests in Nashville, Tennessee, used findings from this case study and travel time, parking availability, and speed data from the local transportation network as inputs to the model found similar results and support the model's generalizability (Cherry, Azad, Rose, & MacArthur, 2019). Mixed fleets of e-cargo cycles and vans can alter associated costs while avoiding heavy congestion in urban cores. In urban cores, replacing vans or trucks with e-trikes can significantly reduce emissions and energy costs. Moreover, using low-emission vehicles, as a branding strategy, could compensate for the relatively small added costs associated with e-trike use in some scenarios.

Urban delivery with e-cargo cycles can take many forms. Among the factors determining cargo cycles classification and selection are the volume, required range, and delivery purpose. Furthermore, vehicle size impacts system effectiveness. In accordance with local requirements and policies, e-trikes may use bicycle infrastructures rather than roads when possible, reducing vehicular congestion and parking constraints. With a micro hub or consolidation center located near the urban core, an e-trike-based delivery system could enhance performance in urban core areas with a high customer density and heavy traffic.

Additionally, cities should consider standard cargo cycle sizes when assessing bike infrastructure policies related to lane widths, access, parking. For instance, the maximum width of the e-trikes in this case study (50 inches) is only 10 inches less than the minimum width of a bike lane proposed by the Federal Highway Administration (FHWA) and wider than some bike lanes, which can be as narrow as 36 inches. As a result, cities should consider multiple uses for bike lanes within urban cores, including cargo delivery.

There are several limitations to this study. First, a single operator in a

single city using one type of vehicle provided all data. That company continues to experiment with other vehicle form factors to improve efficiency. While the proposed methods can apply to other systems, they should be generalized with caution. Portland, Oregon maintains a well-developed bike infrastructure compared to most U.S. cities, less bike-friendly infrastructures may diminish e-trike delivery performance. Other limitations to this study are due to the small sample size, especially in terms of the number of e-trikes included at the time of data collection, and the fact that the data was collected in 2017 and before the COVID-19 pandemic started, which can limit the generalizability of the results. Nevertheless, this case study provides insight into a sustainable delivery service and outlines methods and data sources for developing a sustainable last-mile delivery framework and identifying the stakeholders and policies impacting the implementation of green and competitive delivery systems within urban cores. Additionally, GPS data collection involves two additional limitations: possible data loss due to GPS recorder power failures and poor data accuracy due to suroundings. Future research can overcome this issue with advanced, embedded GPS recorders.

Future studies should also evaluate the operational benefits of existing bike infrastructure and assess operators' cargo cycle delivery system compatibility across multiple vehicle sizes and factors such as bike infrastructure connectivity and accessibility, slope changes in transportation infrastructure, bike lane and sidewalk width, land use, and signal timing. Such studies create fine spatial city maps and help identify potential consolidation center or micro-hub locations, bicycle transportation networks, and public spaces for deliveries. Moreover, e-trike rider safety remains a concern. Qualitative and quantitative studies assessing road safety and transportation conditions in urban cores and providing findings to the public through media, workshops, and other means should improve safety for e-trike riders. Additionally, further studies should evaluate safety using data from various resources such as police crash data or Emergency Medical Systems (EMS) related to urban delivery vehicles, to determine key factors that contribute to accidents.

## Funding

This study was supported, in part, through the State Planning and Research (SPR) Program by the Tennessee Department of Transportation and the Federal Highway Administration under RES #: 2016–31. Partial funding was provided by the USDOT University Transportation Center (UTC) program through the National Institute of Transportation and Communities (NITC).

## Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

The data that has been used is confidential.

## Acknowledgements

The authors thank Franklin Jones at B-line along with the many staff members and riders that participated in the research. We appreciate the support of Kanghao He for data preparation. The authors also appreciate the support provided by the Portland Bureau of Transportation.

## References

- Al Mashalah, H., Hassini, E., Gunasekaran, A., & Bhatt, D. (2022). The impact of digital transformation on supply chains through e-commerce: Literature review and a conceptual framework. *Transportation Research Part E: Logistics and Transportation Review*, 165, Article 102837.
- Arnold, F., Cardenas, I., Sørensen, K., & Dewulf, W. (2018). Simulation of B2C e-commerce distribution in Antwerp using cargo bikes and delivery points. *European transport research review*, 10(1), 2.
- Assmann, T., Lang, S., Müller, F., & Schenk, M. (2020). Impact assessment model for the implementation of cargo bike transshipment points in urban districts. *Sustainability*, 12(10), 4082.
- Austrian Mobility Research. (2017). *Cyclelogistic- moving Europe forward*, Final Public Report Accessed on July 27, 2019. Retrieved from [http://one.cyclelogistics.eu/docs/119/D6\\_9\\_FPR\\_Cyclelogistics\\_print\\_single\\_pages\\_final.pdf](http://one.cyclelogistics.eu/docs/119/D6_9_FPR_Cyclelogistics_print_single_pages_final.pdf).
- B-Line. B-Line Rider Cost. Accessed on Aug 29, 2019. Retrieved from [https://b-linepdx.com/wp-content/uploads/B-line\\_Rider-Warehousing-Ambassador\\_Job\\_Description.pdf](https://b-linepdx.com/wp-content/uploads/B-line_Rider-Warehousing-Ambassador_Job_Description.pdf).
- Balm, S., Moolenburgh, E., Anand, N. and Ploos van Amstel, W. (2018). The Potential of Light Electric Vehicles for Specific Freight Flows: Insights from the Netherlands. In *City Logistics 2* (eds E. Taniguchi and R.G. Thompson). <https://doi.org/10.1002/9781119425526.ch15>.
- Behnke, M. (2019). Recent trends in last mile delivery: impacts of fast fulfillment, parcel lockers, electric or autonomous vehicles, and more. *Logistics Management* (pp. 141–156). Springer.
- Blazejewski, L., Sherrieff, G., & Davies, N. (2020). Delivering the last mile: Scoping the potential for E-cargo bikes. *Project Report* Accessed on June 6, 2021. Retrieved from <http://usir.salford.ac.uk/id/eprint/59007/>.
- Caggiani, L., Prencipe, L. P., Colovic, A., & Dell'Orco, M. (2020). An eco-friendly Decision Support System for last-mile delivery using e-cargo bikes. In *Paper presented at the 2020 IEEE International Conference on Environment and Electrical Engineering and 2020. IEEE Industrial and Commercial Power Systems Europe (IEEEIC/ICPS Europe)*.
- Cairns, S., & Sloman, L. (2019). Potential for e-cargo bikes to reduce congestion and pollution from vans in cities. Accessed on June 6, 2021. Retrieved from <https://www.cistoustopou.cz/sites/default/files/article/2020-11/potential-for-e-cargo-bikes-to-reduce-congestion-and-pollution-from-vans-final.pdf>.
- Chen, X., & Kelly, T. F. (2015). B-Corps—A growing form of social enterprise: Tracing their progress and assessing their performance. *Journal of Leadership and Organizational Studies*, 22(1), 102–114.
- Cherry, C. R., Azad, M., Rose, W. J., & MacArthur, J. (2019). *Alternative Vehicles for Last Mile Freight*. Retrieved from <https://rosap.nrl.bts.gov/view/dot/55072>.
- City Changer Cargo Bike. (2019). CityChangerCargoBike - D3.3 Set of training materials for start-ups, empowerment and multipliers. Accessed on June 5, 2021. Retrieved from [http://cyclelogistics.eu/sites/default/files/downloads/D3.3%20-%20Training%20Materials\\_CityChangerCargoBike\\_final.pdf](http://cyclelogistics.eu/sites/default/files/downloads/D3.3%20-%20Training%20Materials_CityChangerCargoBike_final.pdf).
- Colson, J. R. (2019). *Master's thesis*. Harvard Extension School.
- Conway, A., Cheng, J., Kamga, C., & Wan, D. (2017). Cargo cycles for local delivery in New York City: Performance and impacts. *Research in transportation business & management*, 24, 90–100.
- Dalla Chiara, G., Alho, A. R., Cheng, C., Ben-Akiva, M., & Cheah, L. (2020). Exploring benefits of cargo-cycles versus trucks for urban parcel delivery under different demand scenarios. *Transportation Research Record*, 2674(5), 553–562.
- de Mello Bandeira, R. A., Goes, G. V., Gonçalves, D. N. S., Márcio de Almeida, D. A., & de Oliveira, C. M. (2019). Electric vehicles in the last mile of urban freight transportation: A sustainability assessment of postal deliveries in Rio de Janeiro-Brazil. *Transportation Research Part D: Transport and Environment*, 67, 491–502.
- Dolati Neghabadi, P., Evrard Samuel, K., & Espinouse, M.-L. (2019). Systematic literature review on city logistics: Overview, classification and analysis. *International Journal of Production Research*, 57(3), 865–887.
- Dybdalen, Å., & Ryeng, E. O. (2022). Understanding how to ensure efficient operation of cargo bikes on winter roads. *Research in Transportation Business & Management*, 44, Article 100652.
- Elbert, R., & Friedrich, C. (2020). Urban consolidation and cargo bikes: A simulation study. *Transportation research procedia*, 48, 439–451.
- Electricity Local. (2019). *Portland Commercial Electricity Rates* Accessed on Aug 29 Retrieved from <https://www.electricitylocal.com/states/oregon/portland>.
- Elkington, J. (1998). *Accounting for the triple bottom line, Measuring Business Excellence*, 2 (3), 18–22. <https://doi.org/10.1108/eb025539>.
- Feng, W., & Figliozzi, M. (2013). An economic and technological analysis of the key factors affecting the competitiveness of electric commercial vehicles: A case study from the USA market. *Transportation Research Part C: Emerging Technologies*, 26, 135–145.
- Geroliminis, N. (2015). Cruising-for-parking in congested cities with an MFD representation. *Economics of Transportation*, 4(3), 156–165.
- Giglio, C., Musmanno, R., & Palmieri, R. (2021). Cycle logistics projects in Europe: Intertwining bike-related success factors and region-specific public policies with economic results. *Applied Sciences*, 11(4), 1578.
- Giordano, A., Matthews, H. S., Baptista, P., & Fischbeck, P. (2022). Impacts of topography and weather barriers on commercial cargo bicycle energy using urban delivery crowdsourced cycling data. *Sustainable Cities and Society*, 76, Article 103326.
- Graser, A., & Dragschnig, M. (2020). Exploring movement data in notebook environments. In *Paper presented at the IEEE VIS 2020 Workshop on Information Visualization of Geospatial Networks, Flows and Movement (MoVis)*.
- Graser, A. (2019). MovingPandas: Efficient structures for movement data in Python. *GIForum*, 1, 54–68.
- Gruber, J., & Narayanan, S. (2019). Travel time differences between cargo cycles and cars in commercial transport operations. *Transportation research record*, 2673(8), 623–637.
- Gruber, J. (2021). *Das E-Lastenrad als Alternative im städtischen Wirtschaftsverkehr. Determinanten der Nutzung eines „neuen alten“ Fahrzeugkonzepts*. Humboldt-Universität zu Berlin. Dissertation.
- Gruber, J., Ehrler, V., & Lenz, B. (2013). Technical potential and user requirements for the implementation of electric cargo bikes in courier logistics services. In *Paper presented at the 13th World Conference on Transport Research*.
- Indeed.com. Portland, OR Average Delivery Driver Pay. Accessed on Aug 29, (2019). Retrieved from <https://www.indeed.com/salaries/Delivery-Driver-Salaries,-Portland-OR>.
- International Energy Agency. (2020). Tracking Transport 2020. Accessed on April 20, 2020. Retrieved from <https://www.iea.org/reports/tracking-transport-2020>.
- International Transport Forum. (2021). *ITF Transport Outlook 2021*. Paris: OECD Publishing. <https://doi.org/10.1787/16826a30-en>
- Iwan, S., Nürnberg, M., Jedliński, M., & Kijewska, K. (2021). Efficiency of light electric vehicles in last mile deliveries—Szczecin case study. *Sustainable Cities and Society*, 74, Article 103167.
- Jaller, M., & Pahwa, A. (2021). *The Sustainability of Alternative Last-Mile Delivery Strategies*. Retrieved from <https://escholarship.org/uc/item/83m7s3kg>.
- Katsela, K., Güneş, Ş., Fried, T., Goodchild, A., & Browne, M. J. S. (2022). Defining urban freight microhubs: A case study analysis. 14(1), 532.
- Kelley Blue Book. (2019). *2017 Ram ProMaster City Fuel Economy* Accessed on Aug 29 Retrieved from <https://www.kbb.com/ram/promaster-city/2017/>.
- Kittelson & Associates Inc.. (2015). Centers and corridors parking occupancy report. *Portland Parking Reports* Accessed on Feb 20, 2020. Retrieved from <https://www.porlandoregon.gov/transportation/article/547701>.
- Koning, M., & Conway, A. (2016). The good impacts of biking for goods: Lessons from Paris city. *Case studies on transport policy*, 4(4), 259–268.
- Law, A. M., Kelton, W. D., & Kelton, W. D. (2000). *Simulation modeling and analysis*, 3. New York: McGraw-Hill.
- Lee, K., Chae, J., & Kim, J. (2019). A Courier service with electric bicycles in an urban area: The case in Seoul. *Sustainability*, 11(5), 1255.
- Lenz, B., & Riehle, E. (2013). Bikes for urban freight? Experience in Europe. *Transportation Research Record*, 2379(1), 39–45.
- Lia, F., Nocerino, R., Bresciani, C., COLORNI VITALE, A., & Luè, A. (2014). Promotion of E-bikes for delivery of goods in European urban areas: An Italian case study. In *Paper presented at the Transport Research Arena (TRA) 5th Conference: Transport Solutions from Research to Deployment*.
- Llorca, C., & Moedel, R. (2021). Assessment of the potential of cargo bikes and electrification for last-mile parcel delivery by means of simulation of urban freight flows. *European transport research review*, 13(1), 1–14.
- MacArthur, J., & Kobel, N. (2014). *Regulations of E-bikes in North America*. NITC-RR-564. Portland, OR: Transportation Research and Education Center (TREC). <https://doi.org/10.15760/trec.163>
- Macharis, C., & Kin, B. (2017). The 4 A's of sustainable city distribution: Innovative solutions and challenges ahead. *International Journal of Sustainable Transportation*, 11 (2), 59–71.
- Madlener, R., & Sunak, Y. (2011). Impacts of urbanization on urban structures and energy demand: What can we learn for urban energy planning and urbanization management? *Sustainable Cities and Society*, 1(1), 45–53.
- Maes, J. (2017). *The potential of cargo bicycle transport as a sustainable solution for urban logistics*. University of Antwerp.
- Marujo, L. G., Goes, G. V., D'Agosto, M. A., Ferreira, A. F., Winkenbach, M., & Bandeira, R. A. (2018). Assessing the sustainability of mobile depots: The case of urban freight distribution in Rio de Janeiro. *Transportation Research Part D: Transport and Environment*, 62, 256–267.

- Mayor of London. (2009). *Cycle freight in London: A scoping study* Accessed on July 27, 2019 Retrieved from <http://content.tfl.gov.uk/cycle-as-freight-may-2009.pdf>.
- Melo, S., & Baptista, P. (2017). Evaluating the impacts of using cargo cycles on urban logistics: Integrating traffic, environmental and operational boundaries. *European transport research review*, 9(2), 30.
- Melo, R. A., Zarruk, D., Melo, M. R. A., & Imports RCurl, X. (2017). *Package 'gmapsdistance'*. The Comprehensive R Archive Network (CRAN).
- Navarro, C., Roca-Riu, M., Furió, S., & Estrada, M. (2016). Designing new models for energy efficiency in urban freight transport for smart cities and its application to the Spanish case. *Transportation research procedia*, 12, 314–324.
- Oliveira, C. M.d., Albergaria De Mello Bandeira, R., Vasconcelos Goes, G., Schmitz Gonçalves, D. N., & D'Agosto, M. D. A. (2017). Sustainable vehicles-based alternatives in last mile distribution of urban freight transport: A systematic literature review. *Sustainability*, 9(8), 1324.
- Pahwa, A., & Jaller, M. (2022). A cost-based comparative analysis of different last-mile strategies for e-commerce delivery. *Transportation Research Part E: Logistics and Transportation Review*, 164, Article 102783.
- Ploos van Amstel, W., Balm, S., Warmerdam, J., Boerema, M., Altenburg, M., Rieck, F., & Peters, T. (2018). Urban technology research programme LEFV-logic: Research on light electric freight vehicles. Accessed on August 21, 2020. Retrieved from <http://www.hva.nl/binaries/content/assets/subsites/kc-techniek/publicaties/lef-v-logic.english.pdf>.
- Portland Bureau of Transportation PBOT. (2019). Bicycles in Portland fact sheet. Accessed on July 28, 2019. Retrieved from <https://www.portlandoregon.gov/transportation/article/407660>.
- Quak, H.J. (2008, March 20). Sustainability of Urban Freight Transport: Retail Distribution and Local Regulations in Cities (No. EPS-2008-124-LIS). ERIM Ph.D. Series Research in Management. Erasmus Research Institute of Management. Retrieved from <http://hdl.handle.net/1765/11990>.
- Ranieri, L., Digiesi, S., Silvestri, B., & Roccotelli, M. J. S. (2018). A review of last mile logistics innovations in an externalities cost reduction vision. 10(3), 782.
- Ren, J., Zhao, Q., Liu, B., & Chen, C. (2019). Selection of pallet management strategies from the perspective of supply chain cost with anylogic software. *PLoS one*, 14(6), Article e0217995.
- Robichet, A., Nierat, P., & Combes, F. First and Last Miles by Cargo bikes: ecological commitment or economically feasible? The case of a parcel service company in Paris. *Transportation Research Record*, 03611981221086632.
- Rudolph, C., & Gruber, J. (2017). Cargo cycles in commercial transport: Potentials, constraints, and recommendations. *Research in transportation business & management*, 24, 26–36.
- Rudolph, C., Nsamzinshuti, A., Bonsu, S., Ndiaye, A. B., & Rigo, N. (2022). Localization of relevant urban micro-consolidation centers for last-mile cargo bike delivery based on real demand data and city characteristics. *Transportation Research Record*, 2676 (1), 365–375.
- Russo, S. M., Voegl, J., & Hirsch, P. (2021). A multi-method approach to design urban logistics hubs for cooperative use. *Sustainable Cities and Society*, 69, Article 102847.
- Sadhu, S. S., Tiwari, G., & Jain, H. (2014). Impact of cycle rickshaw trolley (CRT) as non-motorised freight transport in Delhi. *Transport Policy*, 35, 64–70.
- Saenz, J., Figliozzi, M., & Faulin, J. (2016). Assessment of the carbon footprint reductions of tricycle logistics services. *Transportation Research Record*, 2570(1), 48–56.
- Schünemann, J., Finke, S., Severengiz, S., Schelte, N., & Gandhi, S. (2022). Life Cycle Assessment on Electric Cargo Bikes for the Use-Case of Urban Freight Transportation in Ghana. *Procedia CIRP*, 105, 721–726.
- Sheth, M., Butrina, P., Goodchild, A., & McCormack, E. (2019). Measuring delivery route cost trade-offs between electric-assist cargo bicycles and delivery trucks in dense urban areas. *European transport research review*, 11(1), 1–12.
- SmartWay. (2018). *2018 SmartWay Truck Carrier Partner Tool: Technical Documentation U. S. Version 2.0.17 (Data Year 2017)* Accessed on Aug 29, 2019. Retrieved from <https://www.epa.gov/sites/production/files/2018-01/documents/420b18004.pdf>.
- Sun, L., Chen, J., Li, Q., & Huang, D. (2020). Dramatic uneven urbanization of large cities throughout the world in recent decades. *Nature communications*, 11(1), 1–9.
- Taniguchi, E., Thompson, R. G., & Yamada, T. (2016). New opportunities and challenges for city logistics. *Transportation research procedia*, 12, 5–13.
- Tipagornwong, C., & Figliozzi, M. (2014). Analysis of competitiveness of freight tricycle delivery services in urban areas. *Transportation Research Record*, 2410(1), 76–84.
- US Energy Information Administration. (2019). *West Coast (Except California) Average Diesel Prices for the week of 10/10/2017* Accessed on Aug 29 Retrieved from <http://www.eia.gov/petroleum/gasdiesel/>.
- Verlinde, S., Macharis, C., Milan, L., & Kin, B. (2014). Does a mobile depot make urban deliveries faster, more sustainable and more economically viable: Results of a pilot test in Brussels. *Transportation research procedia*, 4, 361–373.
- Wang, J., Parajuli, S., Cherry, C. R., McDonald, N. C., & Lyons, T. (2022). Vulnerable road user safety and freight vehicles: A case study in North Carolina and Tennessee. *Transportation Research Interdisciplinary Perspectives*, 15, Article 100650.
- Wrighton, S., & Reiter, K. (2016). CycleLogistics—moving Europe forward! *Transportation research procedia*, 12, 950–958.
- Zlatanovska, B., Stojkovic, N., Kocaleva, M., Stojanova, A., Lazarova, L., & Gobubovski, R. (2018). Modeling of some chaotic systems with anylogic software. *TEM Journal*, 7(2), 465–470.